



SRI SHAKTHI

INSTITUTE OF ENGINEERING AND TECHNOLOGY

COIMBATORE – 62

Autonomous Institution, Accredited by NAAC with “A” Grade



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21AM422 JAVA PROGRAMMING

LABORATORY

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND MACHINE LEARNING**

**SRI SHAKTHI INSTITUTE OF ENGINEERING AND
TECHNOLOGY**

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AND MACHINE LEARNING

21AM422 JAVA PROGRAMMING

LABORATORY

LABORATORY RECORD

NAME: _____ **ROLLNO:** _____

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INTERNAL EXAMINER

EXTERNAL EXAMINER

INDEX

EXP. NO.	DATE	NAME OF THE EXPERIMENT	MARKS	SIGNATURE
1A	31.12.25	DATATYPE RANGES		
1B	31.12.25	SIMPLE INTEREST CALCULATION		
2	31.12.25	OPERATOR PRECEDENCE		
3A	08.01.26	TERNARY & ASSIGNMENT OPERATOR		
3B	08.01.26	LOGICAL & BITWISE OPERATOR		
4A	23.01.26	STRONG NUMBER		
4B	23.01.26	ARMSTRONG NUMBER		
5	23.01.26	ARRAYS		
6A	23.01.26	SECOND LARGEST NUMBER IN 2D-ARRAY		
6B	23.01.26	SUM OF ALL ELEMENTS IN 2D-ARRAY		
7	19.02.26	INHERITANCE		
8	19.02.26	POLYMORPHISM		
9	19.02.26	ENCAPSULATION		
10	19.02.26	ABSTRACTION		
11	19.02.26	INTERFACE		

SIGNATURE OF THE FACULTY

EXP. NO.	DATE	NAME OF THE EXPERIMENT	MARKS	SIGNATURE
12	05.03.26	COUNT VOWELS IN STRING USING METHODS		
13A	19.03.26	STRING COMPARISON		
13B	19.03.26	CAPITALIZE THE ALTERNATE WORDS		
14A	26.03.26	FRQUENCY COUNT USING HASHMAP		
14B	26.03.26	EMPLOYEE MANAGEMENT USING ARRAY LIST		
15	02.04.26	BANK ACCOUNT TRANSACTION USING EXCEPTION HANDLING		
16	18.04.26	STUDENT MANAGEMENT SYSTEM USING JDBC		
AVERAGE:				

SIGNATURE OF THE FACULTY

EXP.NO:16	STUDENT MANAGEMENT SYSTEM USING JAVA (JDBC) AND MYSQL
DATE:	

AIM:

To develop a Student Management System using Java Swing and JDBC that performs CRUD operations by connecting to a MySQL database.

PROCEDURE:

1. Install Java JDK, Eclipse IDE, and MySQL Workbench.
2. Open MySQL Workbench and create a database named studentdb.
3. Create a table students with fields id (auto increment), name, age, and course.
4. Open Eclipse and create a new Java project.
5. Add the MySQL Connector JAR file to the project build path.
6. Create Java classes: MainClass, LoginForm, StudentForm, and ViewStudents.
7. Design the login form and student form using Java Swing components.
8. Establish database connection using JDBC (DriverManager.getConnection).
9. Implement CRUD operations:
 - a. Insert (Add student)
 - b. Select (View students)
 - c. Update (Modify data)
 - d. Delete (Remove record)
10. Display student records using JTable in the ViewStudents class.
11. Run the application and verify all operations are working correctly.

PROGRAM:

#DATABASE CREATION USING MYSQL

```
CREATE DATABASE studentdb;
USE studentdb;

CREATE TABLE students (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    age INT,
    course VARCHAR(50) )
```

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```

=====
Press:
    1 => Add expense
    2 => View expense
    3 => Update expense
    4 => Delete expense
    5 => Exit
=====
1
Enter amount: 200
Enter category: food
Enter date (YYYY-MM-DD): 2015-08-11
☑ Expense Added!
=====
Press:
    1 => Add expense
    2 => View expense
    3 => Update expense
    4 => Delete expense
    5 => Exit
=====
2

--- Expenses ---
1 | 200.0 | food | 2015-08-11
=====
Press:
    1 => Add expense
    2 => View expense
    3 => Update expense
    4 => Delete expense
    5 => Exit
=====
3
Enter ID to update: 1
Press 1 => Update Amount
Press 2 => Update Category
Press 3 => Update Date
1
Enter new amount: 300
🔪 Updated!
=====

```


#MainClass.java

```
package sms;
public class MainClass {
    public static void main(String[] args) {
        new LoginForm();
    }
}
```

#LoginForm.java

```
package sms;
import javax.swing.*;
import java.awt.event.*;

public class LoginForm extends JFrame implements ActionListener {
    JTextField userField;
    JPasswordField passField;
    JButton loginBtn;

    public LoginForm() {
        setTitle("Login");
        setSize(300, 200);
        setLayout(null);

        JLabel l1 = new JLabel("Username:");
        l1.setBounds(20, 30, 80, 25);
        add(l1);

        userField = new JTextField();
        userField.setBounds(110, 30, 120, 25);
        add(userField);

        JLabel l2 = new JLabel("Password:");
        l2.setBounds(20, 70, 80, 25);
        add(l2);

        passField = new JPasswordField();
        passField.setBounds(110, 70, 120, 25);
        add(passField);

        loginBtn = new JButton("Login");
        loginBtn.setBounds(90, 110, 100, 30);
        loginBtn.addActionListener(this);
        add(loginBtn);

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        String user = userField.getText();
        String pass = new String(passField.getPassword());
    }
}
```

```

=====
Press:
    1 => Add expense
    2 => View expense
    3 => Update expense
    4 => Delete expense
    5 => Exit
=====
2
--- Expenses ---
1 | 300.0 | food | 2015-08-11
=====
Press:
    1 => Add expense
    2 => View expense
    3 => Update expense
    4 => Delete expense
    5 => Exit
=====
4
Enter ID to delete: 1
X Deleted!
=====
Press:
    1 => Add expense
    2 => View expense
    3 => Update expense
    4 => Delete expense
    5 => Exit
=====
2
--- Expenses ---
=====
Press:
    1 => Add expense
    2 => View expense
    3 => Update expense
    4 => Delete expense
    5 => Exit
=====

```

```

if (user.equals("DHANU") && pass.equals("1234")) {
    new StudentForm();
    dispose();
} else {
    JOptionPane.showMessageDialog(this, "Invalid Login");
}
}
}

```

#StudentForm.java

Package sms;

```

import javax.swing.*;
import java.awt.event.*;
import java.sql.*;
public class StudentForm extends JFrame implements ActionListener {

```

```

    JTextField idField, nameField, ageField, courseField;
    JButton addBtn, viewBtn, updateBtn, deleteBtn;

```

```

// SINGLE VIEW WINDOW OBJECT
ViewStudents viewObj;

```

```

public StudentForm() {
    setTitle("Student Form");
    setSize(350, 350);
    setLayout(null);

```

```

// INIT VIEW OBJECT
viewObj = new ViewStudents();

```

```

// ID
JLabel l0 = new JLabel("ID:");
l0.setBounds(20, 20, 80, 25);
add(l0);

```

```

idField = new JTextField();
idField.setBounds(100, 20, 150, 25);
add(idField);

```

```

// Name
JLabel l1 = new JLabel("Name:");
l1.setBounds(20, 60, 80, 25);
add(l1);

```

```

nameField = new JTextField();
nameField.setBounds(100, 60, 150, 25);
add(nameField);

```

```

// Age
JLabel l2 = new JLabel("Age:");
l2.setBounds(20, 100, 80, 25);

```



```

add(l2);

ageField = new JTextField();
ageField.setBounds(100, 100, 150, 25);
add(ageField);

// Course
JLabel l3 = new JLabel("Course:");
l3.setBounds(20, 140, 80, 25);
add(l3);

courseField = new JTextField();
courseField.setBounds(100, 140, 150, 25);
add(courseField);

// Buttons
addBtn = new JButton("Add");
addBtn.setBounds(20, 200, 80, 30);
addBtn.addActionListener(this);
add(addBtn);

viewBtn = new JButton("View");
viewBtn.setBounds(110, 200, 80, 30);
viewBtn.addActionListener(this);
add(viewBtn);

updateBtn = new JButton("Update");
updateBtn.setBounds(200, 200, 100, 30);
updateBtn.addActionListener(this);
add(updateBtn);

deleteBtn = new JButton("Delete");
deleteBtn.setBounds(110, 240, 80, 30);
deleteBtn.addActionListener(this);
add(deleteBtn);

setVisible(true);
}

public void actionPerformed(ActionEvent e) {

    try {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/studentdb",
            "root",
            "Dhanu@24"
        );

        // + ADD
        if (e.getSource() == addBtn) {
            String query = "INSERT INTO students(name, age, course) VALUES(?,?,?)";
            PreparedStatement ps = con.prepareStatement(query);

```



```

ps.setString(1, nameField.getText());
ps.setInt(2, Integer.parseInt(ageField.getText()));
ps.setString(3, courseField.getText());

ps.executeUpdate();
JOptionPane.showMessageDialog(this, "Added!");

// CLEAR FIELDS
idField.setText("");
nameField.setText("");
ageField.setText("");
courseField.setText("");
nameField.requestFocus();

// AUTO REFRESH
viewObj.loadData();
}

// 👁 VIEW
if (e.getSource() == viewBtn) {
    viewObj.setVisible(true);
    viewObj.loadData();
}

// 🔄 UPDATE
if (e.getSource() == updateBtn) {
    String query = "UPDATE students SET name=?, age=?, course=? WHERE id=?";
    PreparedStatement ps = con.prepareStatement(query);

    ps.setString(1, nameField.getText());
    ps.setInt(2, Integer.parseInt(ageField.getText()));
    ps.setString(3, courseField.getText());
    ps.setInt(4, Integer.parseInt(idField.getText()));

    ps.executeUpdate();
    JOptionPane.showMessageDialog(this, "Updated!");

    // AUTO REFRESH
    viewObj.loadData();
}

// ➕ DELETE
if (e.getSource() == deleteBtn) {
    String query = "DELETE FROM students WHERE id=?";
    PreparedStatement ps = con.prepareStatement(query);

    ps.setInt(1, Integer.parseInt(idField.getText()));
    ps.executeUpdate();

    JOptionPane.showMessageDialog(this, "Deleted!");

    // AUTO REFRESH

```



```

        viewObj.loadData();
    }

    con.close();

} catch (Exception ex) {
    System.out.println("Error: " + ex);
}
}
}

```

#ViewStudent.java

```

package sms;
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.sql.*;

public class ViewStudents extends JFrame {

    JTable table;
    DefaultTableModel model;

    public ViewStudents() {
        setTitle("Students List");
        setSize(500, 300);
        setLayout(null);

        // Table columns
        String[] columns = {"ID", "Name", "Age", "Course"};
        model = new DefaultTableModel(columns, 0);

        table = new JTable(model);

        JScrollPane sp = new JScrollPane(table);
        sp.setBounds(20, 20, 440, 200);
        add(sp);

        setVisible(false); // IMPORTANT (we control visibility from StudentForm)
    }

    // LOAD / REFRESH DATA METHOD
    public void loadData() {
        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/studentdb",
                "root",
                "Dhanu@24"
            );

            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM students");

            // Clear old data

```



```

model.setRowCount(0);

// Add fresh data
while (rs.next()) {
    model.addRow(new Object[]{
        rs.getInt("id"),
        rs.getString("name"),
        rs.getInt("age"),
        rs.getString("course")
    });
}

con.close();

} catch (Exception e) {
    System.out.println("Error: " + e);
}
}

```

PROGRAM & EXECUTION	
CLASS PERFORMANCE	
VIVA	
TOTAL	

RESULT:

The Student Management System was successfully developed using Java Swing and MySQL. The application allows users to add, view, update, and delete student records through a graphical interface. The system ensures real-time interaction with the database using JDBC.

